Course Name: Linear Algebra & Calculus
Course Code: MA-111
Course Type: Institute Core

Contact Hours/Week: 3L **Course Credits:** 03

Course Objectives

- To understand concepts of matrix algebra and its applicability in different engineering fields.
- To understand the meaning of the derivative in terms of a rate of change and should be able to use derivatives to solve a variety of problems.
- To able to model a written description of a physical situation with a function, a differential equation, or an integral.
- To have the idea of vector calculus, fundamental theorems & its physical interpretation and applications.
- To develop an appreciation of calculus as a coherent body of knowledge and as a human accomplishment.
- To able to communicate mathematics both orally and in well- written sentences and should be able to explain solutions to problems.

Unit Number	Course Content	Contact Hours
Unit-01	**Introduction to Linear Algebra:** Matrices, Equivalent Matrix, Elementary Matrix, Normal form of a matrix, Gauss-Jordan reduction and inverse of matrices, Row-reduced matrix, Linear dependence and independence of vectors, Rank of a matrix, Consistency and Solution of linear system of equations,	05L
Unit-02	**Eigenvalues and Eigenvectors:** Characteristic equation, Eigen-values, Eigen vectors, Properties of Eigen-values, Orthogonal vectors and its properties, Cayley-Hamilton theorem and its applications.	03L
Unit-03	**Differential Calculus:** Function of two variables, Limit, Continuity and Differentiability, Partial Differentiation and its geometrical interpretation, Homogeneous functions, Euler's theorem and its extension, Total differentials, Composite function, Jacobian, Taylor's and Maclaurin's series (for one and two variables), Maxima and minima of functions of two variables, Method of undetermined multipliers, Curve tracing.	08L
Unit-04	**Integral Calculus:** Double Integrals (Cartesian and Polar), Change of Order of Integration, Change of Variables, Applications of Double Integrals, Triple integrals, Change of variables, Applications of Triple Integrals.	08L
Unit-05	**Vector Calculus (Differential):** Point functions, Differentiation of vectors, General rule of differentiation, Space curves (curves in space), Tangent, Principal normal, Binormal, Osculating plane, Normal plane, Rectifying plane, Curvature and Torsion, Radius of curvature, Frenet's formulae, Tangential and Normal Acceleration, Relative Velocity and Acceleration. Gradient, Divergence and Curl and their Physical Interpretation, Directional derivative, Del applied twice to point function, Del applied to products of point functions.	08L
Unit-06	**Vector Calculus (Integral):** Line Integral, Surface Integral, Volume integrals, Theorems of Green, Stokes and Gauss (without proofs) and their verifications and applications, Irrotational fields, Solenoidal fields	04L

Course Outcomes

Upon successful completion of the course, the student will be able to

CO1: Understand and analyze the theoretical & practical aspects of matrices and calculus.

CO2: Solve systems of linear equations using multiple methods, demonstrate understanding of linear independence,and determine eigenvalues and eigenvectors and solve eigenvalue problems.

CO3: Understand the various solution techniques and practical aspects of differential and Integral calculus.

CO4: Familiar understanding with the basic concepts of differential geometry and the application of vector differentiation and integration to solve the real time surface, area and volume problems.

CO5: Apply the concepts of matrices and calculus in various engineering problems.

Text Books:

1. E. Kreyszig, Advanced Engineering Mathematics, John Wiley and Sons, NC, New York.
2. R.K. Jain and S.R.K. Iyenger, Advanced Engineering Mathematics, Narosa Pub. House.
3. B.V. Ramana, Higher Engineering Mathematics, Mc Graw Hill, India.

Reference Books:

1. G.B. Thomas, M.D. Weir, J. Hass, Thomas' Calculus, Pearson Education India.
2. Gilbert Strang, Calculus, Wellesley-Cambridge Press.
3. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers.